AIROSMITH ENGINEERING

Platform Seismic Analysis Report

October 23, 2025

AT&T FA #:	15412877
AT&T Site Name:	Liverpool BOCES
AT&T PTN #:	2151A0WTHG
Airosmith Project ID:	ATT NSB UNY 202008
Site Location	NYSTA Exit 37 Village of Liverpool, NY 13088 Onondaga County 43° 06′ 05.22″ N NAD83 76° 11′ 11.94″ W NAD83
Demand-Capacity Ratio (CSR)	**23.6%**
Overall Result	**Pass**

PREPARED FOR:

AT&T

STATE OF NEW YORK
JOSEPH R. JOHNSTON
091187
LICENSED PROFESSIONAL ENGINEER
10/23/2025

APPROVED BY: Joseph R. Johnston, P.E.
NY License #: 091187, EXP: 12/31/26
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1.0 Scope

Airosmith Engineering has been requested to perform an analysis of the proposed equipment platform at grade for seismic forces for AT&T Mobility's installation. The platform will support a 3-bay walk-up-cabinet and a 30 kW diesel generator with a 145 gallon belly tank. The platform was analyzed using RISA-3D Version 22.0.1 analysis software. Selected output from the analysis is included in this report.

2.0 Supporting Documentation

Platform Fabrication Drawings	Provided by EMI Drawing #1000-0010-0128, dated 12/26/2018
Construction Drawings	Airosmith Project ID: ATT NSB UNY 202008, Rev 0, dated 3/10/2025

3.0 Analysis Code Requirements

Adopted Building Code	2020 Building Code of NY State / 2018 IBC
Design Standard	ASCE 7-16
Risk Category	II
Exposure Category	C
Topographic Factor Procedure	Method 1, Category 1
Spectral Response	S_s = 0.142 g / S_1 = 0.051 g
Site Class	D – Stiff Soil (Assumed)
Ground Elevation	421.92 ft. (AMSL)

4.0 Results and Conclusions

Upon reviewing the results of this analysis, it is our opinion that the proposed structure meets the specified code requirements. <u>The proposed equipment platform and supporting structure are considered acceptable to support the final loading configuration for seismic forces</u> as listed within in this report. The controlling structure usage is 23.6%.

We appreciate the opportunity to be of service on this project. If you have any questions, require additional information, or actual conditions differ from those as detailed in this report, please contact me via the information below:

Brian Davies, EI
engineering@airosmithdevelopment.com

5.0 Assumptions & Limitations

The following assumptions have been made for this analysis:

- Structural calculations are completed assuming all information provided to Airosmith Engineering is accurate and applicable to this site.
- The existing structures were designed, manufactured, and constructed in accordance with the applicable codes and standards in effect at that time
- The existing structures have been properly maintained in accordance with industry standards.
- All structural and foundation elements, unless otherwise noted, are in good condition, and are capable of supporting their original design capacity.
- Steel grades have been assumed as follows, unless otherwise noted

- Channel, Solid Round, Angle & Plate	ASTM A36 Gr. 36
- HSS (Rectangular)	ASTM A500 Gr. B
- HSS (Pipe)	ASTM A53 Gr. B
- Threaded Rods	ASTM A36 Gr. 36

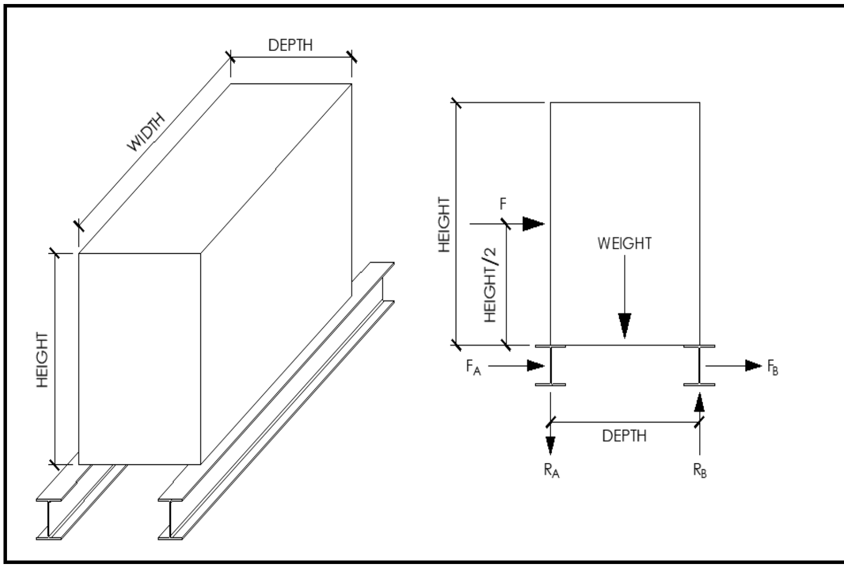
- Calculation-specific assumptions are as noted in the attached appendix

EQUIPMENT PLATFORM LOADS

AIROSMITH ENGINEERING

Airosmith Equipment Platform Loads v2.5

Site Information:	
Date:	10/23/2025
Carrier:	AT&T
Engineer:	DAVI
Site ID:	



Code Information & Wind Data:		
Code Revision:	ASCE7-16	
Basic Wind Speed:		mph
Ice Wind Speed		mph
Ice Thickness		in.
Exposure Category:		
Risk Category:	II	
Topographic Feature:		
Crest Height:		ft.
Lh:		ft.
x:		ft.
Downwind / Upwind		
Ground Elevation:		ft.
K_z:		
K_{zt}:		
K_e:		
K_d:		
G:		
GC_{rh}:		
GC_{rv}:		

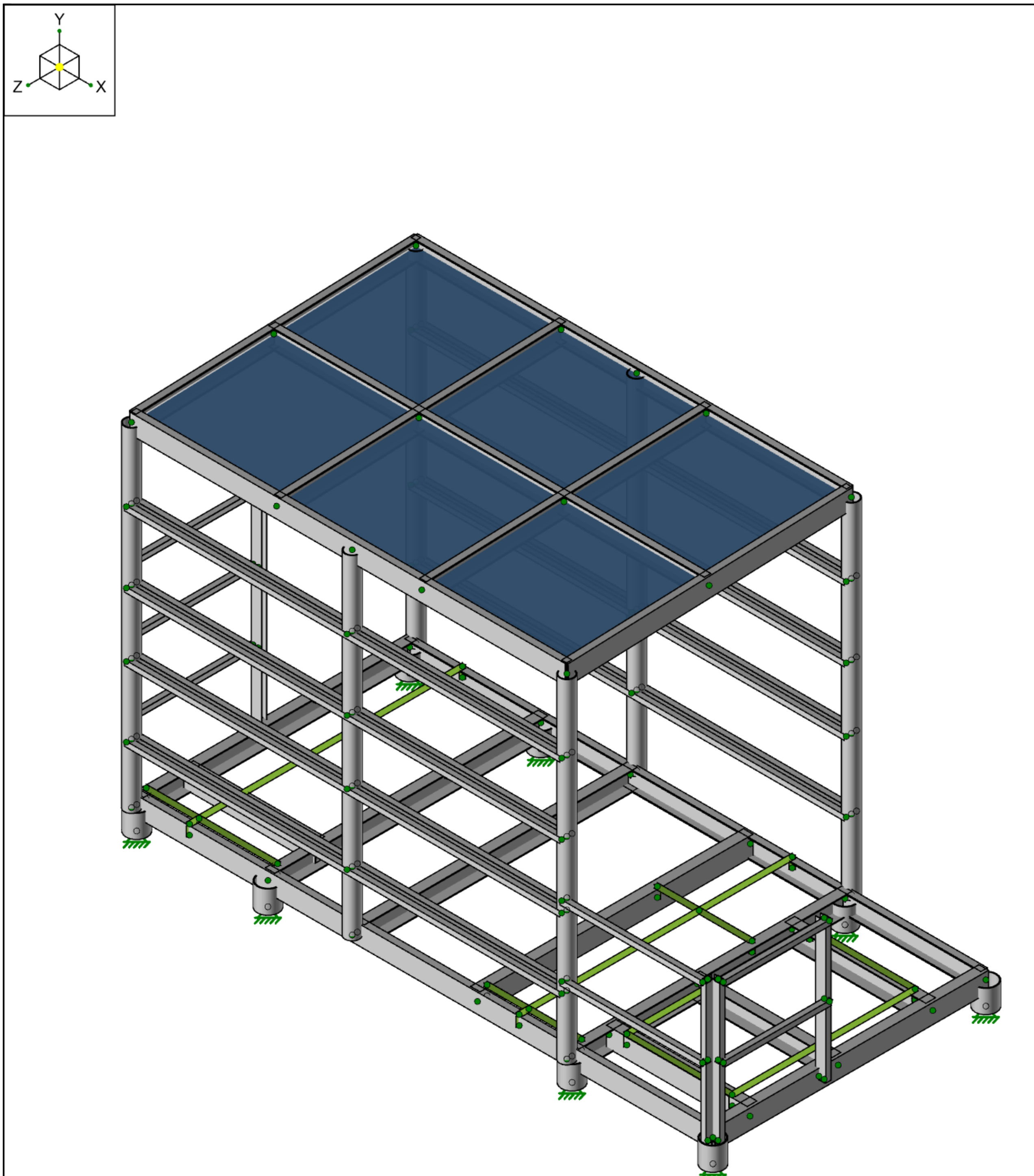
Seismic Data:		
Site Class:	D - Stiff Soil	*Assumed
S_S:	0.142	g
S_1:	0.051	g
F_a:	1.60	
F_v:	2.40	
S_{DS}:	0.15	
S_{D1}:	0.08	
a_p:	1.0	
I_p:	1.0	
R_p:	2.5	
Ω_o:	1.00	

Misc. Dead Loads		
Steel Grating:	10	psf (McNichols GW-125)
Grating Ice Weight:		psf

Design Live Loads		
Table 4.3-1:	20	psf (Roofs - All other construction)

Equipment Loading Information:

Equipment	Base El (ft)	Width (in)	Depth (in)	Height (in)	Weight (lb)
Vertiv 3-Bay Walk-Up-Cabinet (WUC)	1.63	103.00	53.20	74.00	4442
Generac SDC030 w/145 Gal Tank (Full)	1.63	60.70	32.90	109.20	2440



AT&T	Liverpool Boces		SK-1
BrianDavies			Oct 23, 2025 at 02:05 PM
ATT NSB UNY 202008			Platform Design with Unistr…

Hot Rolled Steel Properties

	Label	E [ksi]	G [ksi]	Nu	Therm. Coeff. [1e⁵°F⁻¹]	Density [k/ft³]	Yield [ksi]	Ry	Fu [ksi]	Rt
1	A992	29000	11154	0.3	0.65	0.49	50	1.1	65	1.1
2	A36 Gr.36	29000	11154	0.3	0.65	0.49	36	1.5	58	1.2
3	A572 Gr.50	29000	11154	0.3	0.65	0.49	50	1.1	65	1.1
4	A500 Gr.B RND	29000	11154	0.3	0.65	0.527	42	1.4	58	1.3
5	A500 Gr.B RECT	29000	11154	0.3	0.65	0.527	46	1.4	58	1.3
6	A500 Gr.C RND	29000	11154	0.3	0.65	0.527	46	1.4	62	1.3
7	A500 Gr.C RECT	29000	11154	0.3	0.65	0.527	50	1.4	62	1.3
8	A53 Gr.B	29000	11154	0.3	0.65	0.49	35	1.6	60	1.2
9	A1085	29000	11154	0.3	0.65	0.49	50	1.4	65	1.3
10	A913 Gr.65	29000	11154	0.3	0.65	0.49	65	1.1	80	1.1

Hot Rolled Steel Section Sets

	Label	Shape	Type	Design List	Material	Design Rule	Area [in²]	Iyy [in⁴]	Izz [in⁴]	J [in⁴]
1	C6x10.5	C6X10.5	Beam	Channel	A572 Gr.50	Typical	3.07	0.86	15.1	0.128
2	1.625x3.25 Unistrut	1.625X3.25UNISTRUT	Beam	Channel	A572 Gr.50	Typical	0.821	0.865	0.414	0.003
3	1.625x1.625 Unistrut	1.625X1.625UNISTRUT	Beam	Channel	A572 Gr.50	Typical	0.354	0.099	0.167	0.0006445
4	1/4" PL Support Joist	1/4"PLSUPPORTJOIST	Beam	Channel	A572 Gr.50	Typical	2.875	2.488	15.872	0.057
5	6.0 Pipe	PIPE_6.0	Beam	HSS Pipe	A572 Gr.50	Typical	5.2	26.5	26.5	52.9
6	4.0 Pipe	PIPE_4.0	Beam	HSS Pipe	A572 Gr.50	Typical	2.96	6.82	6.82	13.6

Member Point Loads (BLC 1 : Dead Load)

	Member Label	Direction	Magnitude [lb, lb-ft]	Location [(in, %)]
1	M1	Y	-380	50
2	M1	Y	-380	84
3	M1	Y	-380	118
4	M1	Y	-380	152
5	M2	Y	-380	50
6	M2	Y	-380	84
7	M2	Y	-380	118
8	M2	Y	-380	152

Member Point Loads (BLC 2 : Seismic Load Z)

	Member Label	Direction	Magnitude [lb, lb-ft]	Location [(in, %)]
1	M1	Y	38.52	50
2	M1	Y	12.84	84
3	M1	Y	-12.84	118
4	M1	Y	-38.52	152
5	M2	Y	38.52	50
6	M2	Y	12.84	84
7	M2	Y	-12.84	118
8	M2	Y	-38.52	152
9	M1	Z	-27.63	50
10	M1	Z	-27.63	84
11	M1	Z	-27.63	118
12	M1	Z	-27.63	152
13	M2	Z	-27.63	50
14	M2	Z	-27.63	84
15	M2	Z	-27.63	118
16	M2	Z	-27.63	152

Member Point Loads (BLC 3 : Seismic Load X)

	Member Label	Direction	Magnitude [lb, lb-ft]	Location [(in, %)]
1	M1	Y	41.18	50
2	M1	Y	41.18	84
3	M1	Y	41.18	118
4	M1	Y	41.18	152
5	M2	Y	-41.18	50
6	M2	Y	-41.18	84
7	M2	Y	-41.18	118
8	M2	Y	-41.18	152
9	M1	X	-27.63	50
10	M1	X	-27.63	84
11	M1	X	-27.63	118
12	M1	X	-27.63	152
13	M2	X	-27.63	50
14	M2	X	-27.63	84
15	M2	X	-27.63	118
16	M2	X	-27.63	152

Member Area Loads (BLC 1 : Dead Load)

	Node A	Node B	Node C	Node D	Direction	Load Direction	A Magnitude [psf]	B Magnitude [psf]	C Magnitude [psf]	D Magnitude [psf]	Exclude Braces
1	N2	N1	N11	N12	Y	Two Way	-10	-10	-10	-10	Yes

Member Area Loads (BLC 4 : Live Load)

	Node A	Node B	Node C	Node D	Direction	Load Direction	A Magnitude [psf]	B Magnitude [psf]	C Magnitude [psf]	D Magnitude [psf]	Exclude Braces
1	N2	N1	N11	N12	Y	Two Way	-20	-20	-20	-20	Yes

Basic Load Cases

	BLC Description	Category	X Gravity	Y Gravity	Z Gravity	Point	Distributed	Area(Member)
1	Dead Load	DL		-1		8		1
2	Seismic Load Z	ELZ			-0.073	16		
3	Seismic Load X	ELX	-0.073			16		
4	Live Load	LL						1
5	BLC 1 Transient Area Loads	None					48	
6	BLC 4 Transient Area Loads	None					48	

Load Combinations

	Description	Solve	P-Delta	BLC	Factor	BLC	Factor	BLC	Factor	BLC	Factor
1	(1.2+0.2 Sds)D + 1.0 LL + 1.0 Eh 000 + 0.2 SL	Yes	Y	DL	1.23	LL	1	ELZ	1	SL	0.2
2	(1.2+0.2 Sds)D + 1.0 LL + 1.0 Eh 090 + 0.2 SL	Yes	Y	DL	1.23	LL	1	ELX	1	SL	0.2
3	(0.9-0.2Sds)D + 1.0 Eh 000	Yes	Y	DL	0.87	ELZ	1				
4	(0.9-0.2Sds)D + 1.0 Eh 090	Yes	Y	DL	0.87	ELX	1				

Envelope AISC 15TH (360-16): LRFD Member Steel Code Checks

	Member	Shape	Code Check	Loc[in]	LC	Shear Check	Loc[in]	Dir	LC	phi*Pnc [lb]	phi*Pnt [lb]	phi*Mn y-y [lb-ft]	phi*Mn z-z [lb-ft]	Cb	Eqn
1	M1	C6X10.5	0.236	41.25	1	0.029	135	y	1	5996.437	138150	3372.549	10708.343	1.346	H1-1b
2	M2	C6X10.5	0.233	41.25	2	0.029	41.25	y	2	5996.437	138150	3372.549	10566.651	1.328	H1-1b
3	M147	1.625X1.625UNISTRUT	0.127	69.417	2	0.002	136	y	1	1213.984	15946.875	298.922	712.517	2.733	H1-1b*
4	M125	1.625X1.625UNISTRUT	0.121	136	1	0.002	136	y	1	1213.984	15946.875	298.922	782.132	3	H1-1b*
5	M61	1.625X1.625UNISTRUT	0.113	69.417	2	0.002	136	y	1	1213.984	15946.875	298.922	700.632	2.687	H1-1b*

Envelope AISC 15TH (360-16): LRFD Member Steel Code Checks (Continued)

	Member	Shape	Code Check	Loc[in]	LC	Shear Check	Loc[in]	Dir	LC	phi*Pnc [lb]	phi*Pnt [lb]	phi*Mn y-y [lb-ft]	phi*Mn z-z [lb-ft]	Cb	Eqn
6	M60	1.625X1.625UNISTRUT	0.099	69.417	2	0.002	136	y	1	1213.984	15946.875	565.567	705.921	2.708	H1-1b*
7	M122	PIPE_4.0	0.038	0	2	0.01	0		2	94384.74	133200	15187.5	15187.5	1	H1-1b
8	M13	1/4"PLSUPPORTJOIST	0.037	0	1	0.013	2.664	z	1	70020.922	129375	6177.549	23261.719	3	H1-1b
9	M34	PIPE_4.0	0.036	0	2	0.009	0		2	92307.651	133200	15187.5	15187.5	1	H1-1b
10	M39	PIPE_4.0	0.028	0	1	0.008	0		1	94384.74	133200	15187.5	15187.5	1	H1-1b
11	M58	1.625X1.625UNISTRUT	0.028	0	2	0.002	0	y	2	1213.984	15946.875	298.922	703.196	2.697	H1-1b
12	M149	1.625X1.625UNISTRUT	0.028	0	2	0.002	0	y	2	1213.984	15946.875	298.922	706.395	2.709	H1-1b
13	M35	PIPE_4.0	0.027	0	1	0.006	21.276		2	92307.651	133200	15187.5	15187.5	1	H1-1b
14	M15	PIPE_6.0	0.025	0	2	0.02	7.125		2	233829.625	234000	39750	39750	1	H1-1b
15	M20	PIPE_6.0	0.025	0	2	0.019	7.125		2	233829.625	234000	39750	39750	1	H1-1b
16	M159	C6X10.5	0.023	136	1	0.005	90.667	y	1	10504.139	138150	3372.549	15104.992	1.416	H1-1b
17	M158	C6X10.5	0.023	69.417	1	0.003	90.667	y	2	10504.139	138150	3372.549	15221.39	1.427	H1-1b
18	M57	1.625X1.625UNISTRUT	0.021	0	2	0.002	0	y	2	1213.984	15946.875	298.922	699.65	2.684	H1-1b
19	M16	PIPE_6.0	0.021	0	1	0.016	7.125		1	233829.625	234000	39750	39750	1	H1-1b
20	M153	1.625X1.625UNISTRUT	0.021	0	2	0.002	0	y	2	1213.984	15946.875	298.922	703.49	2.698	H1-1b
21	M4	1/4"PLSUPPORTJOIST	0.021	2.664	1	0.006	55.945	y	1	70020.922	129375	6177.549	23261.719	2.643	H1-1b
22	M118	1.625X1.625UNISTRUT	0.021	0	2	0.002	0	y	2	1213.984	15946.875	298.922	716.768	2.749	H1-1b
23	M33	PIPE_4.0	0.021	107.499	2	0.004	86.224		2	92307.651	133200	15187.5	15187.5	1	H1-1b
24	M19	PIPE_6.0	0.021	0	1	0.015	0.148		1	233829.625	234000	39750	39750	1	H1-1b
25	M127	1.625X1.625UNISTRUT	0.021	0	2	0.002	0	y	2	1213.984	15946.875	298.922	720.053	2.762	H1-1b
26	M124	PIPE_4.0	0.02	104.187	2	0.004	82.481		2	94384.74	133200	15187.5	15187.5	1	H1-1b
27	M59	1.625X1.625UNISTRUT	0.02	0	2	0.002	0	y	2	1213.984	15946.875	565.567	702.189	2.693	H1-1b
28	M148	1.625X1.625UNISTRUT	0.019	0	2	0.002	0	y	2	1213.984	15946.875	565.567	704.862	2.704	H1-1b
29	M117	1.625X1.625UNISTRUT	0.019	0	2	0.002	0	y	2	1213.984	15946.875	565.567	721.886	2.769	H1-1b
30	M56	1.625X1.625UNISTRUT	0.019	0	2	0.002	0	y	2	1213.984	15946.875	565.567	701.42	2.69	H1-1b
31	M5	1/4"PLSUPPORTJOIST	0.019	41.737	1	0.008	85.25	y	1	70020.922	129375	6920.104	22646.607	1.208	H1-1b
32	M131	1.625X1.625UNISTRUT	0.019	0	2	0.002	0	y	2	1213.984	15946.875	565.567	724.956	2.781	H1-1b
33	M152	1.625X1.625UNISTRUT	0.019	0	2	0.002	0	y	2	1213.984	15946.875	565.567	704.969	2.704	H1-1b
34	M6	1/4"PLSUPPORTJOIST	0.015	2.664	2	0.009	2.664	z	1	70020.922	129375	6920.104	23261.719	2.738	H1-1b
35	M163	C6X10.5	0.015	90.667	1	0.001	0	y	2	10504.139	138150	3372.549	12145.032	1.138	H1-1b
36	M10	1/4"PLSUPPORTJOIST	0.013	38.5	2	0.009	38.5	y	1	109856.309	129375	6177.549	23261.719	3	H1-1b
37	M8	1/4"PLSUPPORTJOIST	0.012	85.25	1	0.008	7.992	y	2	70020.922	129375	6920.104	23261.719	2.254	H1-1b
38	M11	1/4"PLSUPPORTJOIST	0.01	38.5	1	0.006	0	y	1	109856.309	129375	6177.549	23261.719	3	H1-1b
39	M38	1.625X3.25UNISTRUT	0.01	0	1	0.002	21.812	y	1	19513.079	36925.875	743.324	2112.294	2.689	H1-1b
40	M80	1.625X1.625UNISTRUT	0.01	44	1	0.002	44	y	1	8822.153	15946.875	298.922	861.372	2.305	H1-1b
41	M160	C6X10.5	0.009	44.468	1	0.002	6.485	y	1	24562.705	138150	3372.549	20919.011	1.33	H1-1b
42	M3	1/4"PLSUPPORTJOIST	0.009	2.664	1	0.009	55.945	y	2	70020.922	129375	6920.104	23261.719	2.673	H1-1b
43	M157	C6X10.5	0.009	44.468	1	0.002	6.485	y	2	24562.705	138150	3372.549	20953.856	1.332	H1-1b
44	M161	C6X10.5	0.007	44.468	1	0.001	88.937	y	1	24562.705	138150	3372.549	18339.33	1.166	H1-1b
45	M162	C6X10.5	0.007	44.468	2	0.001	88.937	y	2	24562.705	138150	3372.549	18342.265	1.166	H1-1b
46	M79	1.625X1.625UNISTRUT	0.007	0	2	0.001	44	y	1	8822.153	15946.875	298.922	861.372	2.773	H1-1b
47	M14	PIPE_6.0	0.006	0	2	0.004	0		2	233829.625	234000	39750	39750	1	H1-1b
48	M17	PIPE_6.0	0.006	0	2	0.005	7.125		2	233829.625	234000	39750	39750	1	H1-1b
49	M18	PIPE_6.0	0.005	0	1	0.004	7.125		1	233829.625	234000	39750	39750	1	H1-1b
50	M21	PIPE_6.0	0.005	0	1	0.004	0		1	233829.625	234000	39750	39750	1	H1-1b
51	M32	1.625X3.25UNISTRUT	0.005	0	2	0.001	31.75	y	2	15687.139	36925.875	2760.212	2112.294	2.037	H1-1b
52	M12	1/4"PLSUPPORTJOIST	0.005	38.5	1	0.006	38.5	y	1	109856.309	129375	6177.549	23261.719	3	H1-1b
53	M36	1.625X3.25UNISTRUT	0.005	0	2	0.001	21.812	y	1	19513.079	36925.875	743.324	2112.294	2.571	H1-1b
54	M70	1.625X1.625UNISTRUT	0.005	0	2	0.001	9.479	y	2	9517.389	15946.875	298.922	861.372	2.452	H1-1b
55	M31	1.625X3.25UNISTRUT	0.004	0	2	0.001	43.656	y	2	15687.139	36925.875	2760.212	2112.294	1.95	H1-1b
56	M37	1.625X3.25UNISTRUT	0.004	0	2	0.001	21.812	y	1	19513.079	36925.875	743.324	2112.294	2.684	H1-1b
57	M9	1/4"PLSUPPORTJOIST	0.004	20.927	1	0.003	0	y	2	112248.385	129375	6920.104	23261.719	1.139	H1-1b
58	M68	1.625X1.625UNISTRUT	0.004	39.563	2	0.001	9.479	y	2	9517.389	15946.875	298.922	861.372	1.981	H1-1b
59	M69	1.625X1.625UNISTRUT	0.003	0	2	0.001	9.479	y	2	9517.389	15946.875	565.567	861.372	2.545	H1-1b
60	M82	1.625X1.625UNISTRUT	0.003	33.125	1	0.001	0	y	2	10466.975	15946.875	298.922	861.372	3	H1-1b

Envelope AISC 15TH (360-16): LRFD Member Steel Code Checks (Continued)

	Member	Shape	Code Check	Loc[in]	LC	Shear Check	Loc[in]	Dir	LC	phi*Pnc [lb]	phi*Pnt [lb]	phi*Mn y-y [lb-ft]	phi*Mn z-z [lb-ft]	Cb	Eqn
61	M81	1.625X1.625UNISTRUT	0.002	33.125	1	0.001	33.125	y	1	10466.975	15946.875	298.922	861.372	1.913	H1-1b

Envelope Node Reactions

	Node Label		X [lb]	LC	Y [lb]	LC	Z [lb]	LC	MX [lb-ft]	LC	MY [lb-ft]	LC	MZ [lb-ft]	LC
1	N28	max	148.694	2	411.63	1	-81.913	3	0	4	0	4	0	4
2		min	17.668	3	171.417	4	-287.08	2	0	1	0	1	0	1
3	N21	max	-206.179	4	499.174	1	-24.592	2	0	4	0	4	0	4
4		min	-260.994	1	262.16	4	-72.977	3	0	1	0	1	0	1
5	N22	max	-173.306	3	519.359	2	87.038	3	0	4	0	4	0	4
6		min	-310.502	2	245.955	3	37.077	2	0	1	0	1	0	1
7	N24	max	1425.874	2	1945.209	2	89.915	3	0	4	0	4	0	4
8		min	782.685	3	1029.453	3	-68.966	2	0	1	0	1	0	1
9	N23	max	1360.246	2	1987.558	2	115.417	1	0	4	0	4	0	4
10		min	723.664	3	1094.662	3	1.329	4	0	1	0	1	0	1
11	N25	max	-506.016	4	2549.763	1	-30.876	2	0	4	0	4	0	4
12		min	-1023.131	1	1482.275	4	-51.725	3	0	1	0	1	0	1
13	N27	max	99.375	2	302.735	1	311.627	1	0	4	0	4	0	4
14		min	-34.334	3	136.568	4	111.815	4	0	1	0	1	0	1
15	N26	max	-554.971	4	2299.446	1	231.417	3	0	4	0	4	0	4
16		min	-1079.93	1	1239.216	4	41.272	2	0	1	0	1	0	1
17	Totals:	max	408.807	4	10357.784	2	408.756	3						
18		min	0.001	1	5818.769	3	-0.002	2						

SLIDING DUE TO SEISMIC FORCES

Maximum Sliding Force:	409 lbs	[Factored]
Maximum Total Downforce:	10,358 lbs	[Factored]
Friction Coefficient:	0.30	[Galv. Steel / Gravel]
Strength Reduction Factor:	0.90	
Design Sliding Resistance:	2,797 lbs	

RESULTS

Maximum FactoredSliding Force < Design Sliding Resistance

PASS

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